MINI PROJECT REPORT

ON

Smart Bus Tracking and Monitoring System

Submitted By

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in partial fulfillment of the requirements for the award of the degree
of

Bachelor of Technology
in

Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COLLEGE OF ENGINEERING, CHERTHALA

ALAPPUZHA-688541

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COLLEGE OF ENGINEERING, CHERTHALA

ALAPPUZHA-688541

C E R T I F I C A T E

This is to certify that, the project report titled **"Smart Bus Tracking and Monitoring System"** is a bonafide record of the **CSD 334 Mini Project** presented by **Akash V G (CEC23CS016), Fidha Fathima A (CEC23CS064), Mohammed Ramshad T S(CEC23CS086), Nithya Manoj (CEC23CS093)**, Sixth semester Bachelor of Technology student, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree **B.Tech in Computer Science and Engineering** of **APJ Abdul Kalam Technological University** during the academic year 2025-2026.

Guide	**Co-ordinator**	**HOD**
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ACKNOWLEDGEMENT

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ABSTRACT

The Smart Bus Tracking and Monitoring System automates student attendance and improves safety in college transportation using a single smartphone. The system applies real-time face recognition to identify students during boarding and exit, thereby reducing manual errors and proxy attendance. Each attendance event is recorded with GPS coordinates and timestamps to ensure accurate tracking of boarding and exit locations. A centralized backend stores student records, attendance logs, and route information, enabling administrators to monitor operations through a web-based dashboard.

The system also detects unregistered individuals and prevents duplicate attendance entries, which improves accountability and security. Because the solution uses only the smartphone camera and GPS, it is cost-effective, easy to deploy, and suitable for small and medium-sized institutions. Overall, the proposed system provides a practical alternative to hardware-intensive transport monitoring approaches while improving safety, accuracy, and administrative efficiency.

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Chapter 1

INTRODUCTION

Student transportation management in educational institutions requires both accurate attendance and reliable safety monitoring. Conventional methods are often manual or depend on additional hardware such as RFID, which can introduce delays, proxy entries, maintenance overhead, and incomplete records during daily bus operations.

To address these limitations, this project proposes a smartphone-based Smart Bus Tracking and Monitoring System that integrates real-time face recognition with GPS tracking. The system automatically records boarding and exit attendance with timestamps and location data, and provides centralized monitoring through an admin dashboard. This approach improves accuracy, transparency, and operational efficiency while remaining cost-effective and easy to deploy.

The system is also designed to support day-to-day administrative decision-making by maintaining structured logs of student movement, attendance events, and route activity. These records can be reviewed to identify delays, monitor service reliability, and verify student travel history when required. Such data-driven visibility reduces dependence on manual verification and strengthens accountability in campus transportation management.

By combining mobile computing, face recognition, and location intelligence in a single workflow, the proposed model provides a practical foundation for modernizing institutional transport operations. The solution is particularly suitable for colleges seeking an affordable and scalable implementation without deploying additional dedicated hardware infrastructure.

Chapter 2

PROBLEM STATEMENT

2.1 Problem Statement

Ensuring student safety and accurate bus attendance is challenging with manual methods and RFID systems due to errors, delays, and hardware dependency, creating a need for a low-cost, automated real-time solution.

2.2 Objective

The primary objectives of the Smart Bus Tracking and Monitoring System are to develop an efficient, automated, and real-time solution for managing student transportation, ensuring safety, accuracy in attendance, and improved monitoring capabilities.

1. **Automated Attendance System:** To design a system that uses facial recognition technology to automatically identify students and record their attendance as they board and exit the bus, eliminating manual errors.

2. **Real-time Bus Tracking:** To implement GPS-based tracking to monitor the live location of the bus, enabling administrators to track routes and ensure timely transportation.

3. **Cost-effective Solution:** To develop a system that operates using a single smartphone without requiring additional hardware, making it affordable and easy to deploy.

4. **Improved Safety and Monitoring:** To enhance student safety by providing accurate records of boarding and exiting along with real-time tracking information.

5. **Reduced Manual Effort:** To minimize human intervention in attendance marking and monitoring, thereby increasing efficiency and reducing delays.

6. **Centralized Data Management:** To store attendance and location data in a centralized system, allowing easy access and monitoring by administrators.

7. **User-friendly Interface:** To create an intuitive platform that enables administrators to easily view attendance records and track bus movements.

Chapter 3

LITERATURE SURVEY

3.1 Paper Review

The paper [1] presents an IoT and GPS-based system for real-time monitoring and tracking of buses. The primary objective of this system is to provide accurate location information to users and reduce waiting time at bus stops. The methodology involves installing GPS modules in buses to continuously transmit location data, which is then displayed through a monitoring interface. The system offers advantages such as improved safety, efficient tracking, and better time management for passengers. However, the system is highly dependent on internet connectivity and GPS accuracy, which may lead to inconsistencies in real-time tracking. Additionally, it does not include any mechanism for identifying passengers or automating attendance, thereby limiting its application in educational transport systems.

The paper [2] focuses on automating attendance using facial recognition techniques. The system captures images of individuals and compares them with a stored database to identify and mark attendance. The methodology includes face detection, feature extraction, and matching using machine learning algorithms. The system offers significant advantages such as contactless attendance, prevention of proxy attendance, and reduced manual effort. However, its performance is affected by environmental factors such as lighting conditions, variations in face angles, and occlusion (e.g., masks or obstacles). Moreover, the system lacks integration with tracking mechanisms, making it unsuitable for applications that require location-based monitoring.

The paper [3] proposes an intelligent system that integrates artificial intelligence with facial recognition and barcode scanning for attendance management. The methodology ensures accurate identification and automated recording of attendance, improving reliability and efficiency. The system provides advantages such as automation, reduced human intervention, and prevention of fraudulent attendance practices. However, the system does not include real-time alert mechanisms or location tracking features. The dependency on multiple identification methods also increases system complexity and implementation cost.

The paper [4] presents an integrated system combining IoT-based GPS tracking with RFID-based attendance monitoring. The system automatically records student entry and exit while providing live tracking of bus movement. The advantages include improved monitoring, automation, and enhanced visibility of transportation activities. However, the system relies on RFID technology, which requires additional hardware and can increase implementation cost. Furthermore, the absence of computer vision techniques such as facial recognition limits its accuracy and flexibility.

The paper [5] introduces a real-time bus tracking system with Estimated Time of Arrival (ETA) prediction. The system uses GPS data and routing algorithms to calculate accurate arrival times, thereby reducing waiting time and improving user convenience. The system enhances transparency and provides better communication between users and transport services. However, it does not include any attendance monitoring system and is highly dependent on network connectivity and GPS precision, which may affect performance in certain environments.

From the above literature survey, it is evident that existing systems either focus on bus tracking or attendance management independently. Many of these systems depend on additional hardware components or suffer from environmental and operational limitations. Therefore, there is a clear need for a unified, cost-effective system that integrates real-time tracking and automated attendance using a single device. The proposed system addresses this gap by combining face recognition and GPS technologies within a smartphone-based platform, ensuring improved accuracy, efficiency, and ease of deployment.

Chapter 4

PROPOSED SYSTEM

The Smart Bus Tracking and Monitoring System is proposed as a cost-effective, real-time transportation monitoring solution that automates attendance and improves student safety using a single smartphone. The system combines face recognition and GPS tracking to record student boarding and exit events accurately without requiring additional hardware.

Functions included are:

1. **Automated Face-based Attendance:** Detects and identifies students as they board and exit the bus, reducing manual effort and proxy attendance.
2. **GPS-based Real-time Tracking:** Captures live bus location and logs accurate boarding/exit coordinates with timestamps.
3. **Single-device Deployment:** Uses only a smartphone camera and GPS, avoiding RFID cards or dedicated hardware.
4. **Centralized Data Storage:** Stores attendance records, trip logs, and route details in a backend database for administrator access.
5. **Duplicate and Unauthorized Detection:** Prevents repeated attendance entries and flags unregistered individuals.
6. **Admin Dashboard Monitoring:** Provides a web-based interface to view attendance history, route activity, and student movement details.
7. **Improved Safety and Accountability:** Ensures transparent and verifiable transport records for students and management.

Chapter 5

SOFTWARE AND HARDWARE REQUIREMENT

5.1 Software Requirements

- **Frontend :** Flutter (Dart) for mobile application development.
- **Backend :** Python Flask for server-side processing.
- **Database :** Firebase Firestore for storing student data, attendance logs, and GPS data.
- **Face Detection :** Google ML Kit for real-time face detection.
- **Face Recognition :** TensorFlow Lite for on-device machine learning.
- **APIs Used :** Google Maps SDK, Geolocation API, Directions API.
- **Development Tools :** Android Studio / VS Code.

5.2 Hardware Requirements

- **Device :** Smartphone with camera and GPS support.
- **Processor :** Minimum Quad-core processor.
- **Main Memory :** 4GB RAM or higher.
- **Storage :** Minimum 64GB.
- **Camera :** Integrated mobile camera (for face recognition).
- **Connectivity :** Internet connection (4G/5G/WiFi) for real-time data sync.

Chapter 6

SYSTEM DESIGNS

6.1 Data Flow Diagram

The data flow diagram (DFD) is used for classifying system requirements to major transformation that will become programs in system design. This is starting point of the design phase that functionally decomposes the required specifications down to the lower level of details
Bubbles: Represent the data transformations.
Lines: Represent the logic flow of data.
Data can trigger events and can be processed to useful information. Systems analysis recognizes the central goal of data in organizations.

6.1.1 Level 0

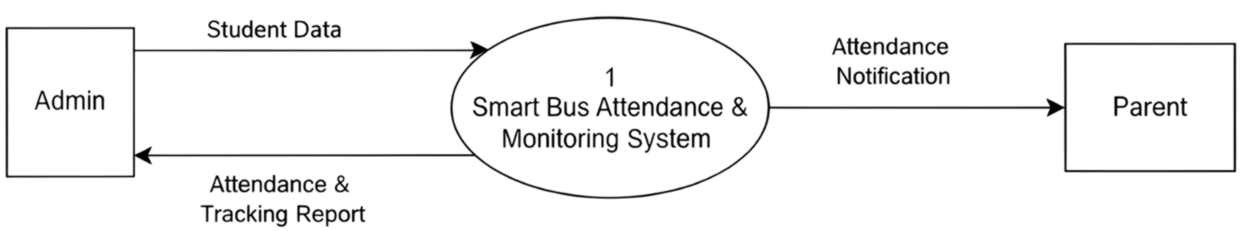

Fig. 6.1: DFD Level 0

6.1.2 Level 1

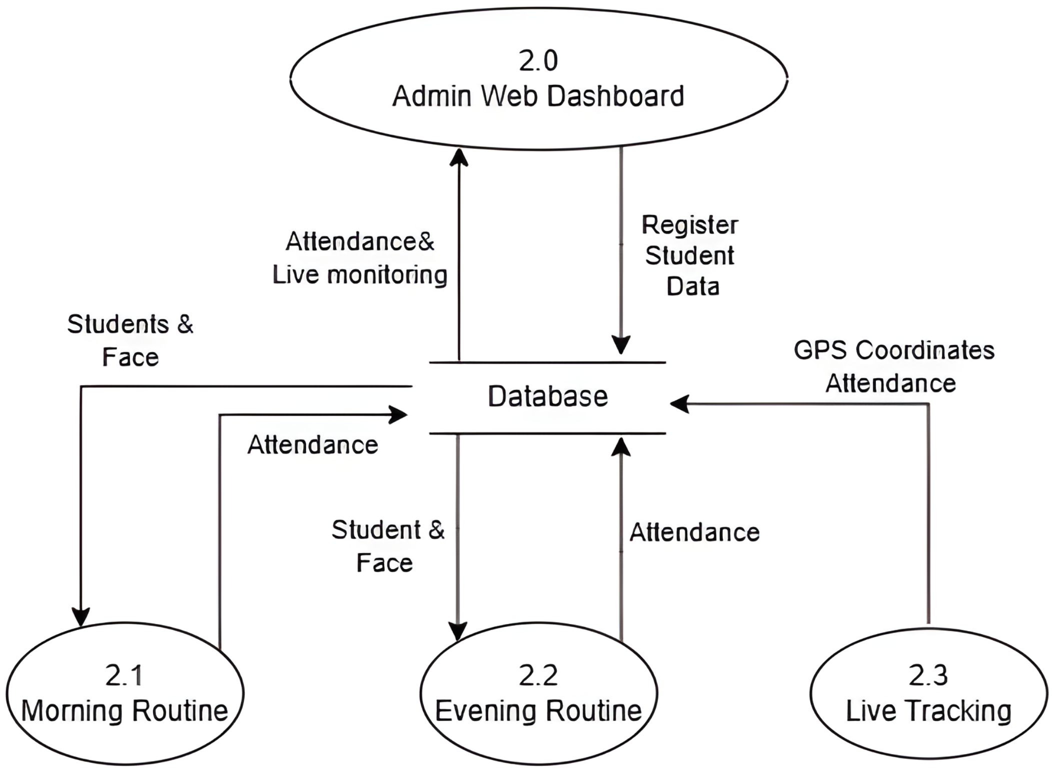

Fig. 6.2: DFD Level 1

6.1.3 Level 2.1

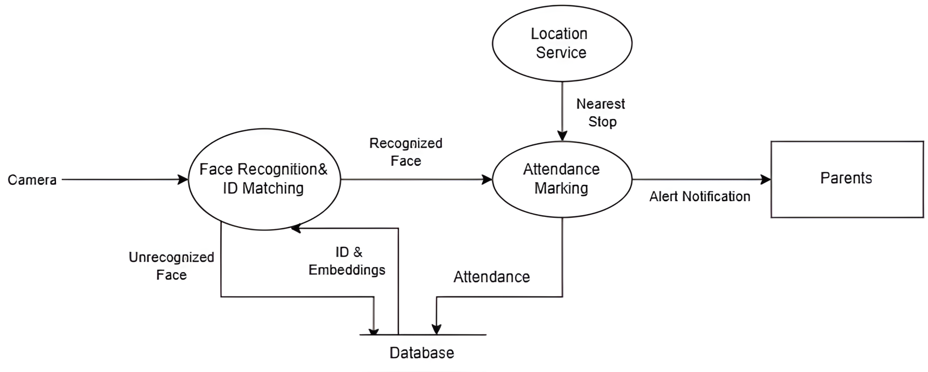

Fig. 6.3: DFD Level 2.1

6.1.4 Level 2.2

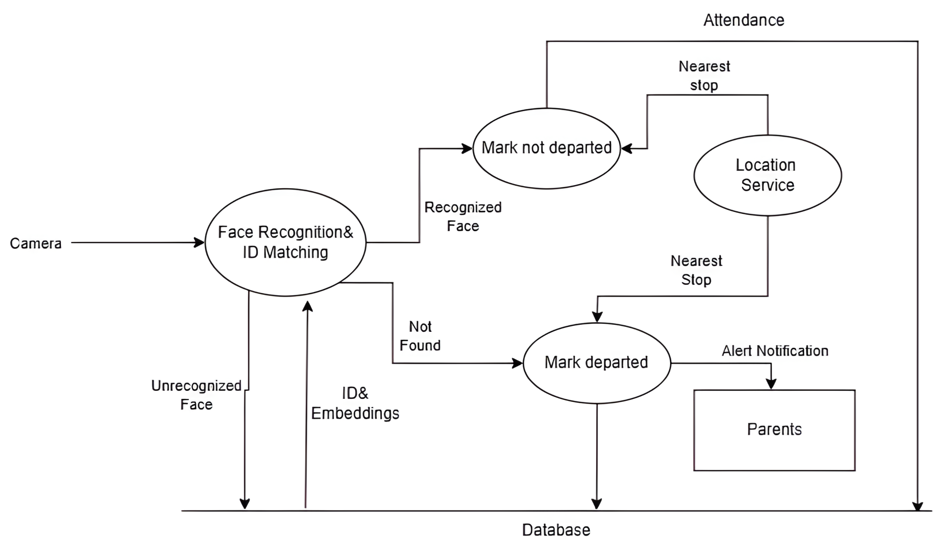

Fig. 6.4: DFD Level 2.2

6.1.5 Level 2.3

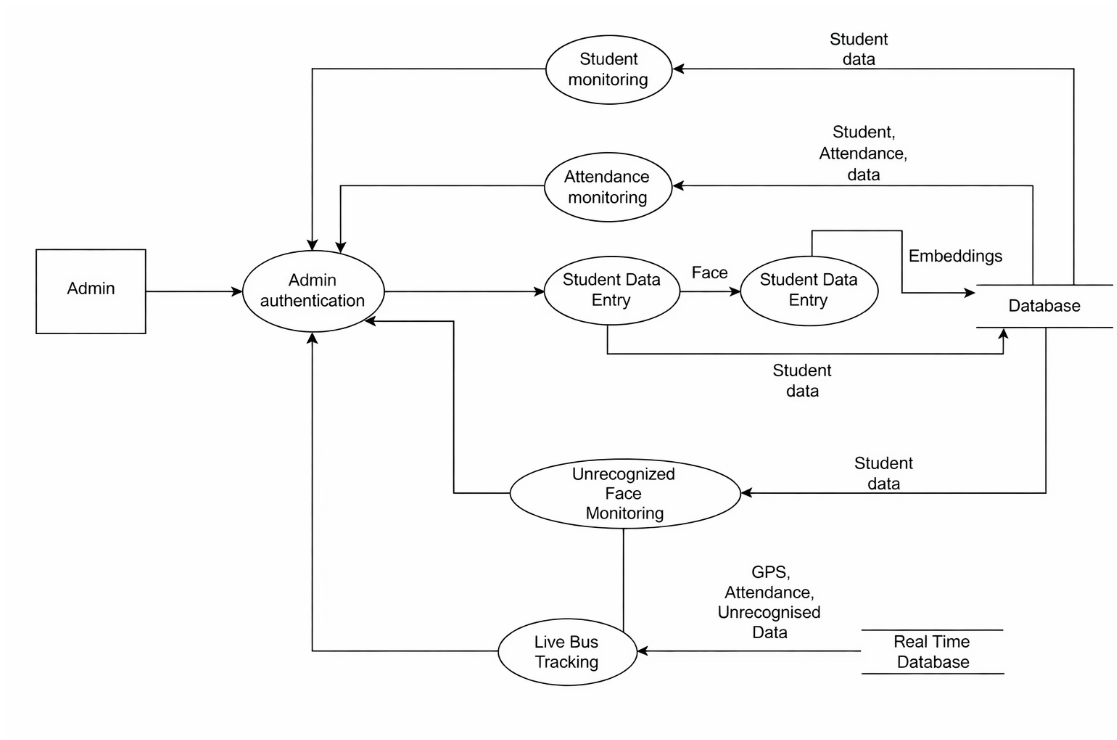

Fig. 6.5: DFD Level 2.3

6.1.6 Level 2.4

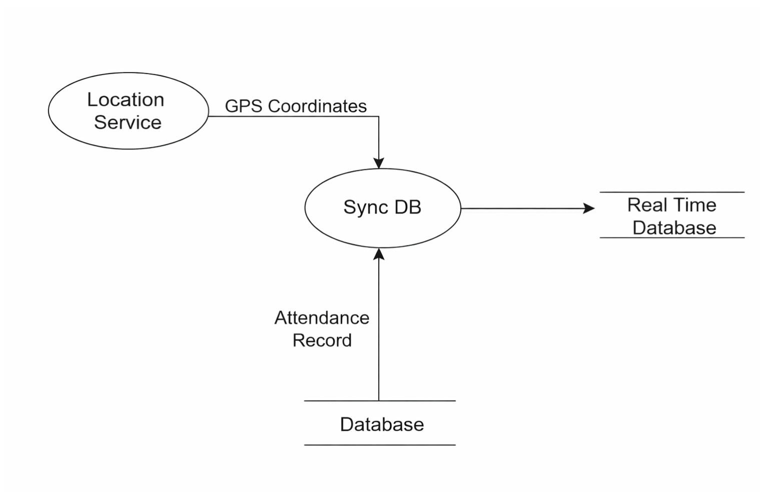

Fig. 6.6: DFD Level 2.4

6.2 System Architecture

The system architecture presents a high-level view of the Smart Bus Tracking and Monitoring System, its major modules, and their interactions. It explains the flow of data from image capture to face recognition, attendance recording, and GPS-based tracking, thereby clarifying the end-to-end system operation.

System Workflow Explanation

The system workflow is divided into three major components: **Face Detection and Feature Extraction**, **Recognition and Processing Engine**, and **Data Storage and Reporting**.

Input and Face Detection Module

This initial phase focuses on capturing student data and extracting facial features for recognition.

- **Image Capture:** The system uses a smartphone camera to capture real-time images of students while boarding and exiting the bus.
- **Face Detection:** Google ML Kit is used to detect faces from the captured image frames efficiently.
- **Feature Extraction:** Facial features are extracted and converted into embeddings using TensorFlow Lite for accurate recognition.
- **Input Feature Vector:** The generated facial embeddings act as feature vectors and are passed to the recognition module.

Recognition and Processing Engine

This phase is responsible for identifying students and processing attendance.

- **Face Matching:** The extracted embeddings are compared with stored embeddings in the database to identify the student.

- **Attendance Marking:** Once a match is found, attendance is automatically recorded with date and time.
- **Duplicate Prevention:** The system ensures that repeated or duplicate entries are avoided.
- **Unrecognized Detection:** If no match is found, the system flags the individual as unregistered.

Tracking and Reporting Module

This phase handles location tracking and data management.

- **GPS Tracking:** The system captures real-time bus location using the device's GPS.
- **Data Storage:** Attendance records, student details, and GPS logs are stored in Firebase Firestore.
- **Admin Monitoring:** Administrators can monitor attendance and live bus location through a web dashboard.
- **Report Generation:** The system provides structured reports of attendance and travel history for analysis and monitoring.

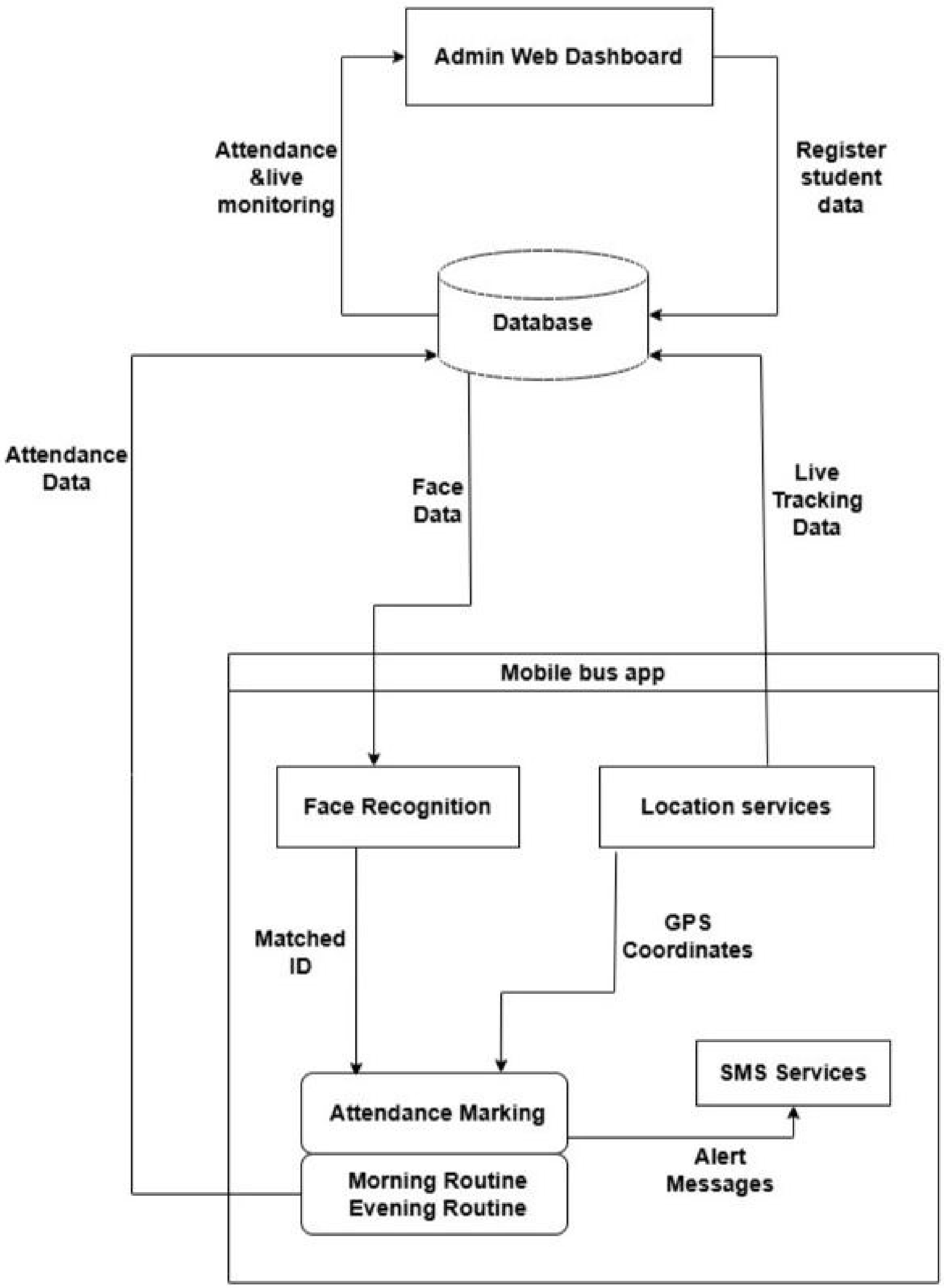

Fig. 6.7: System Architecture

Chapter 7

IMPLEMENTATION AND TESTING

This chapter describes the implementation of the Smart Bus Tracking and Monitoring System. It explains the development environment, tools used, and how each module of the system was implemented to achieve the project objectives.

7.1 Coding Environment Used

The system was developed using modern tools and technologies suitable for mobile and web-based applications. The frontend mobile application was developed using Flutter, which enables cross-platform development with a single codebase. The backend was implemented using Python Flask to handle server-side operations and API integration. Firebase Firestore was used as the database for storing student details, attendance records, and GPS logs.

Face detection was implemented using Google ML Kit, while TensorFlow Lite was used for face recognition through embedding generation and matching. The development was carried out using Visual Studio Code and Android Studio, and all dependencies were managed using standard development tools.

7.2 System Modules Implementation

7.2.1 Image Capture and Input Module

This module is responsible for capturing real-time images of students using the smartphone camera. The application provides an interface for the driver or operator to initiate attendance. The captured images are processed immediately, ensuring quick and efficient handling of input data.

7.2.2 Face Recognition Module

This is the core module of the system, responsible for identifying students and marking attendance. The workflow includes:

- **Face Detection:** Faces are detected from the captured images using Google ML Kit.
- **Feature Extraction:** Facial features are converted into embeddings using TensorFlow Lite.
- **Face Matching:** The generated embeddings are compared with stored embeddings in the database to identify the student.
- **Attendance Marking:** Once a match is found, attendance is automatically recorded with date and time.
- **Duplicate Prevention:** The system ensures that repeated entries are avoided.

7.2.3 Tracking Module

This module handles real-time tracking of the bus location using GPS. The system continuously captures location data and updates it to the database. This enables live tracking of the bus and helps in monitoring boarding and exit points of students.

7.2.4 Database and Reporting Module

This module manages data storage and retrieval. Firebase Firestore is used to store student information, attendance logs, and GPS data. Administrators can view attendance details and track bus location through a web dashboard. The system also generates structured reports for monitoring and analysis.

Functional testing was carried out for key workflows, including student enrollment, face recognition-based attendance, duplicate-entry prevention, and GPS log updates. The observed behavior confirms that the integrated modules operate consistently for real-time monitoring and attendance management.

Chapter 8

RESULT

8.1 Screenshots

8.1.1 Admin login

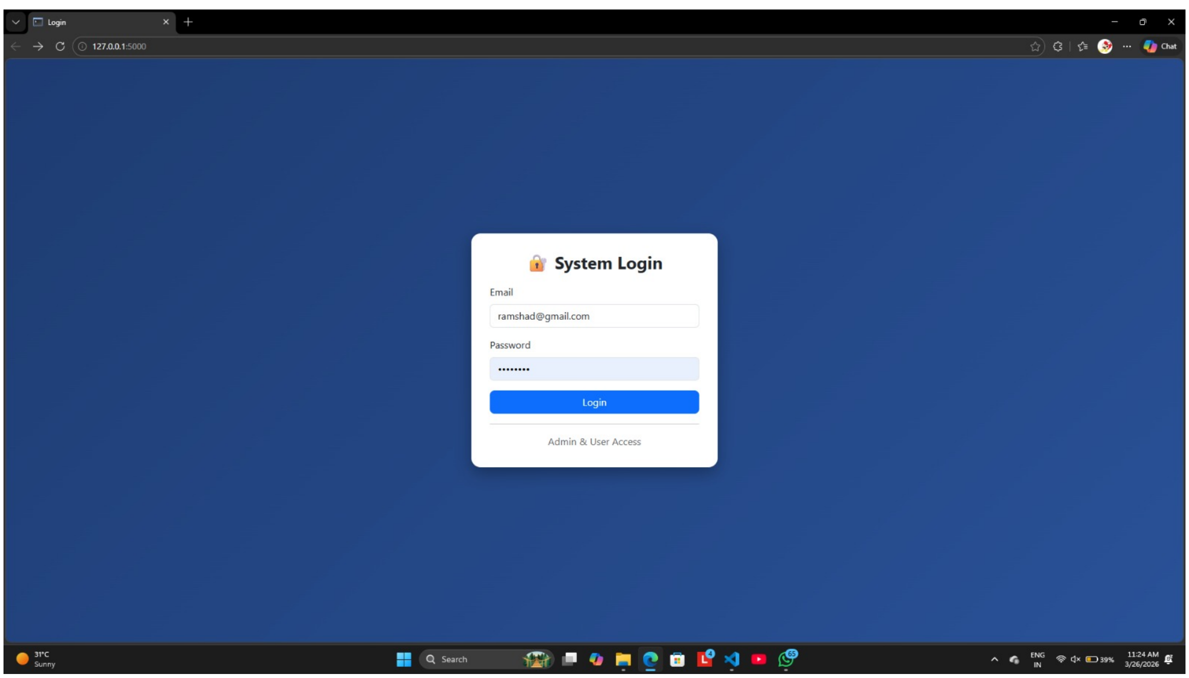

Fig. 8.1: Admin login

8.1.2 Admin Dashboard

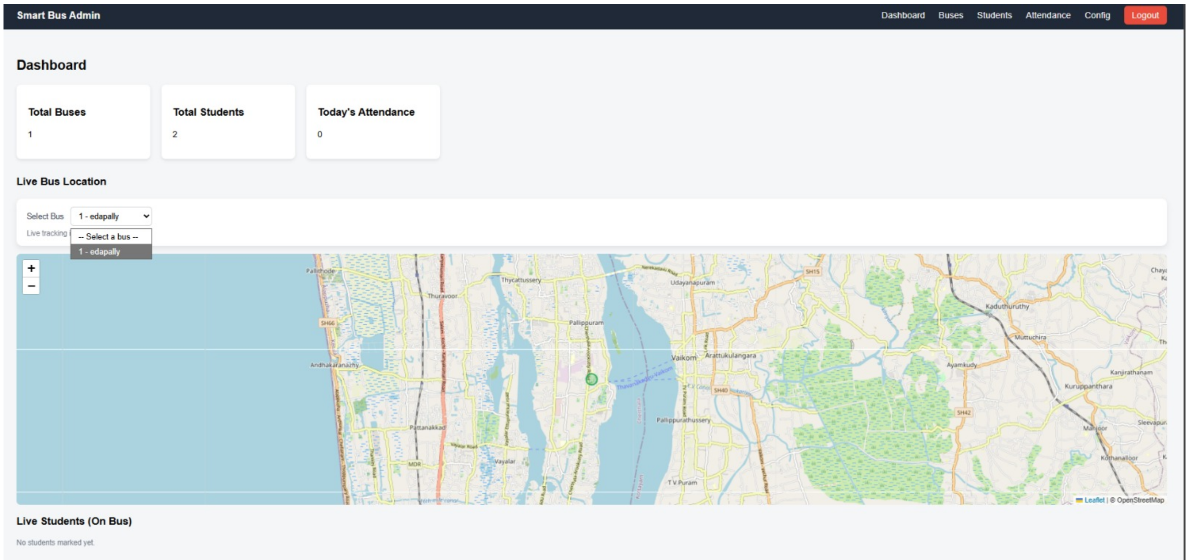

Fig. 8.2: Admin Dashboard

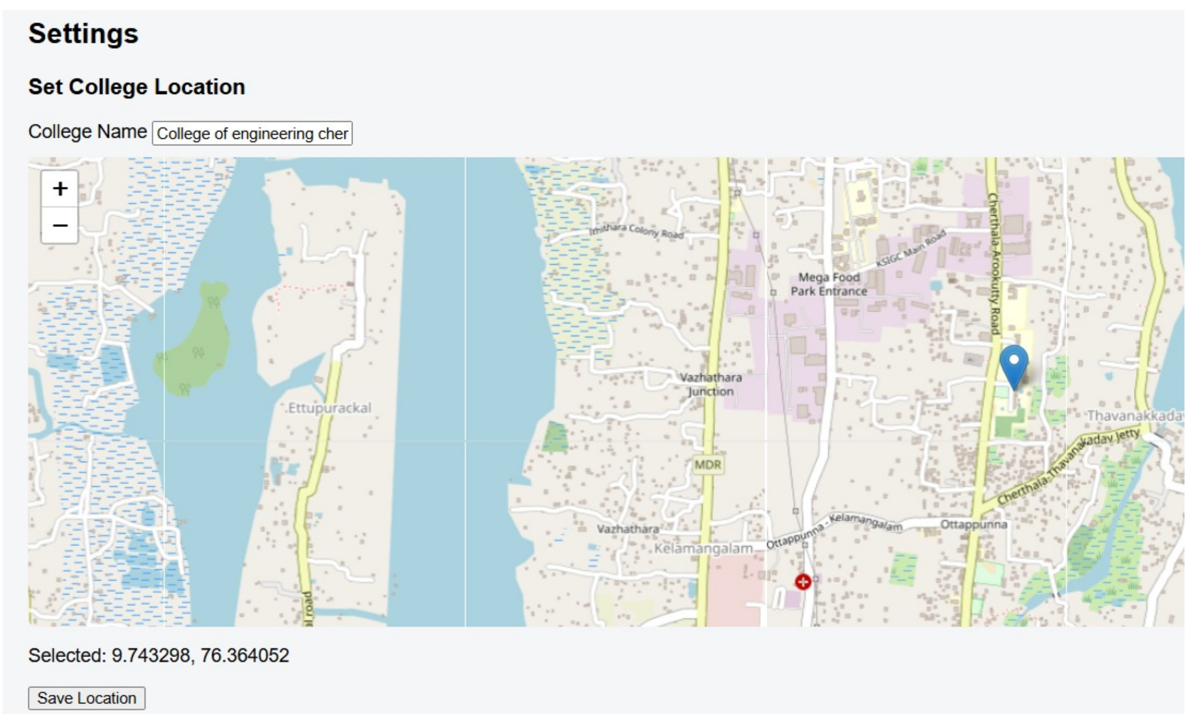

Fig. 8.3: Admin Interface - Add location

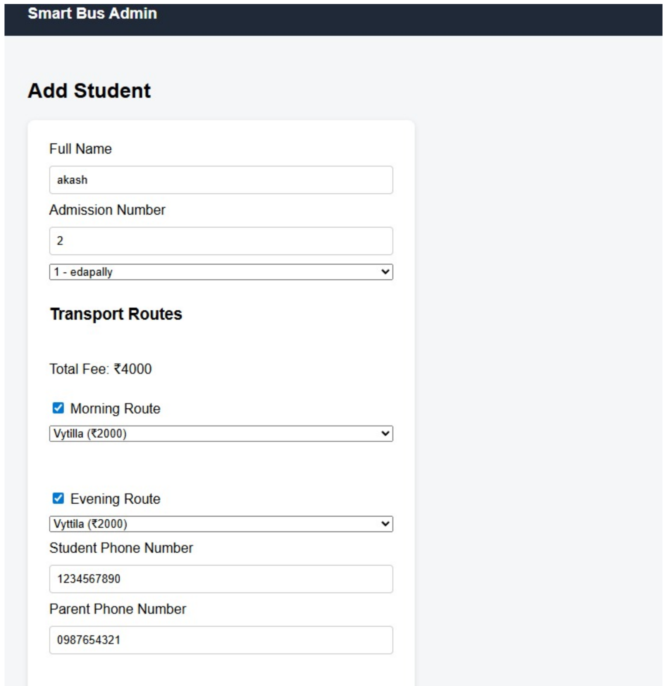

Fig. 8.4: Admin Interface - Add Student

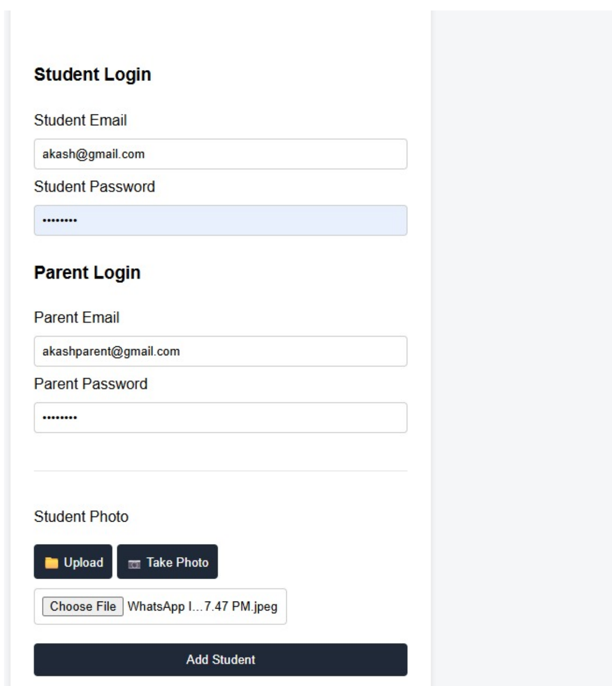

Fig. 8.5: Admin Interface - Student Login

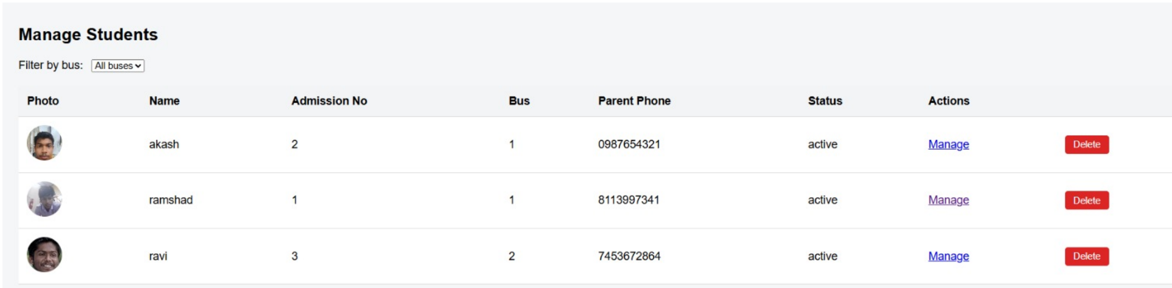

Fig. 8.6: Admin Interface - Manage Student

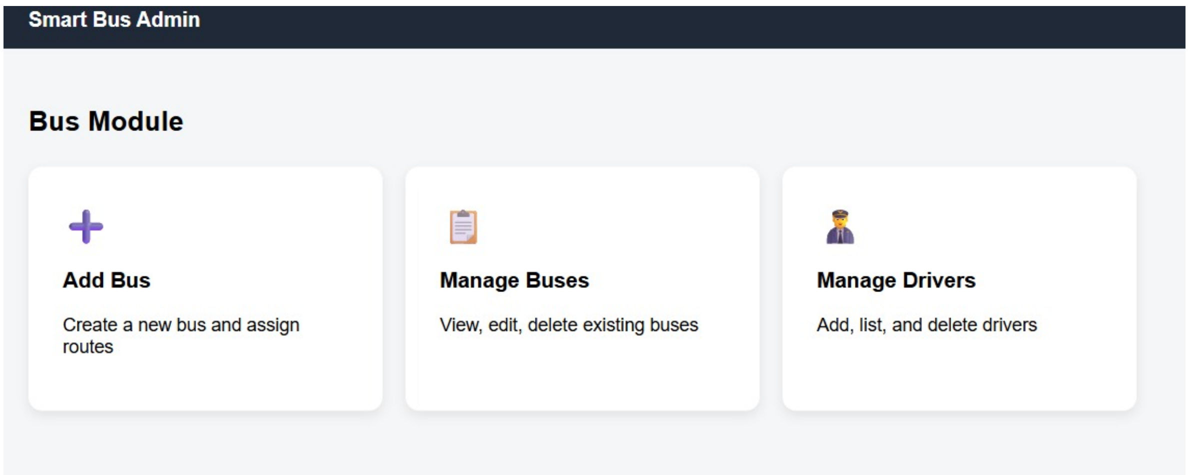

Fig. 8.7: Admin Interface - Bus Module

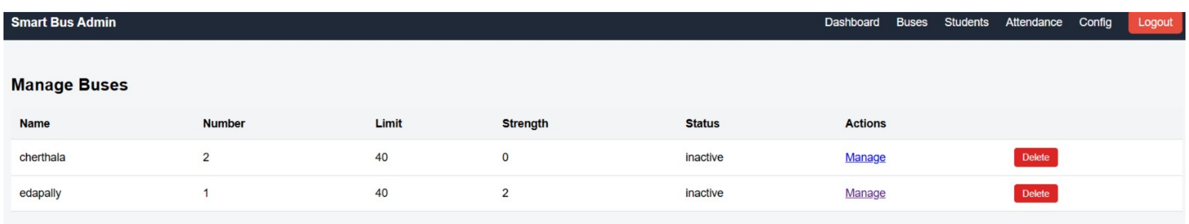

Fig. 8.8: Admin Interface - Manage Buses

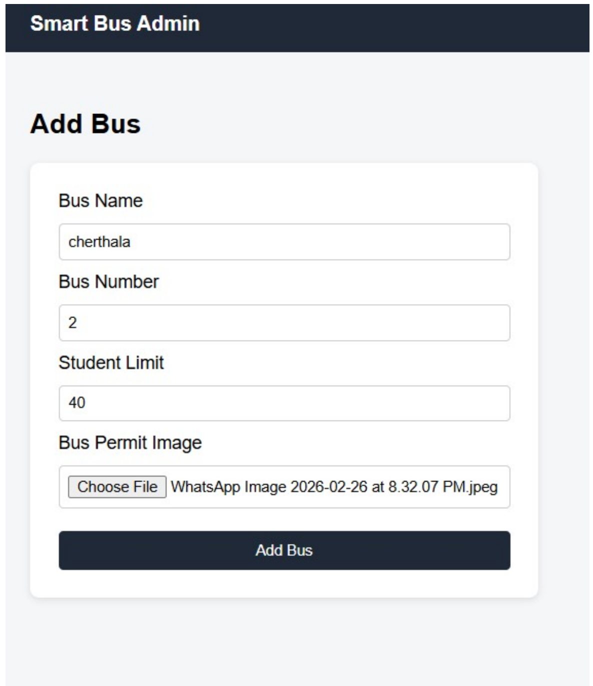

Fig. 8.9: Admin Interface - Add Bus

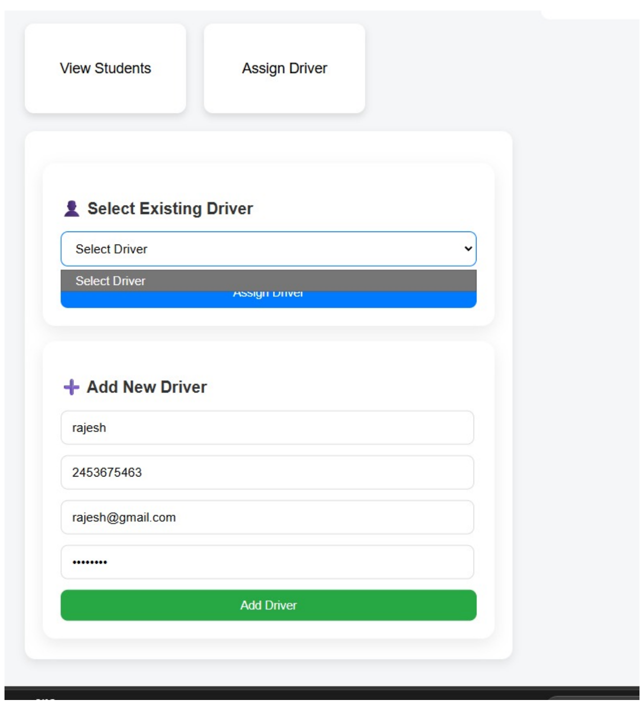

Fig. 8.10: Admin Interface - Choose Driver

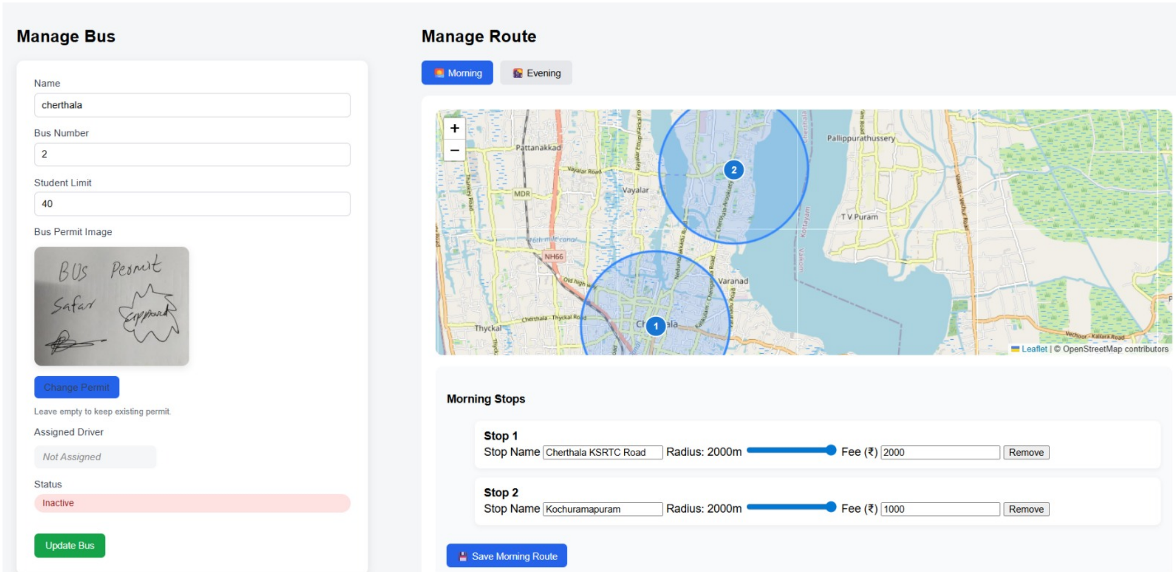

Fig. 8.11: Admin Interface - Bus and Route

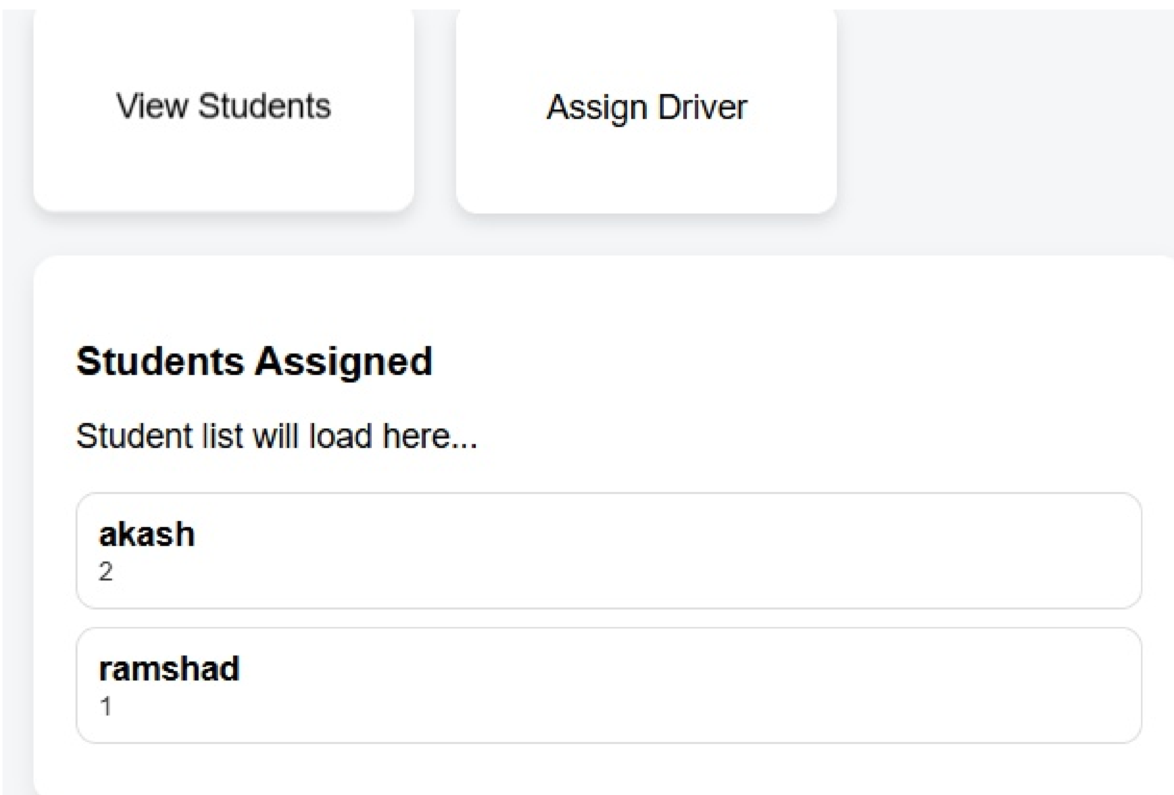

Fig. 8.12: Admin Interface - Students Assigning

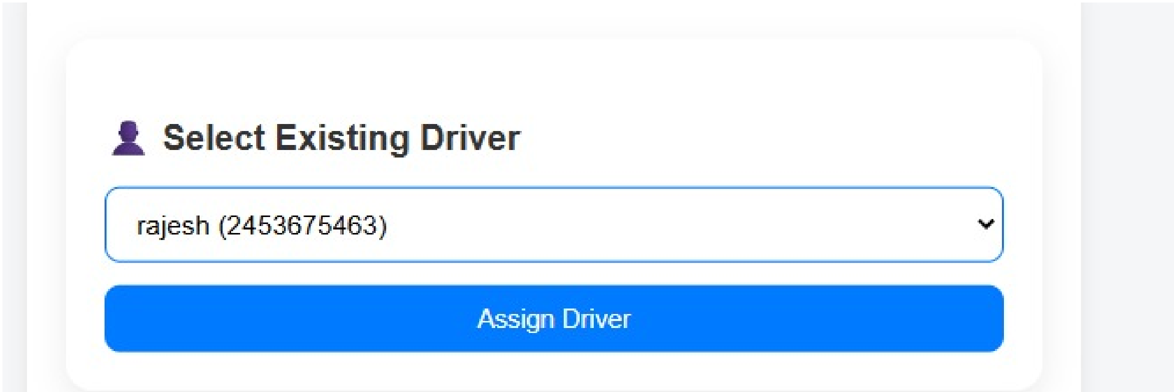

Fig. 8.13: Admin Interface - Assign Driver

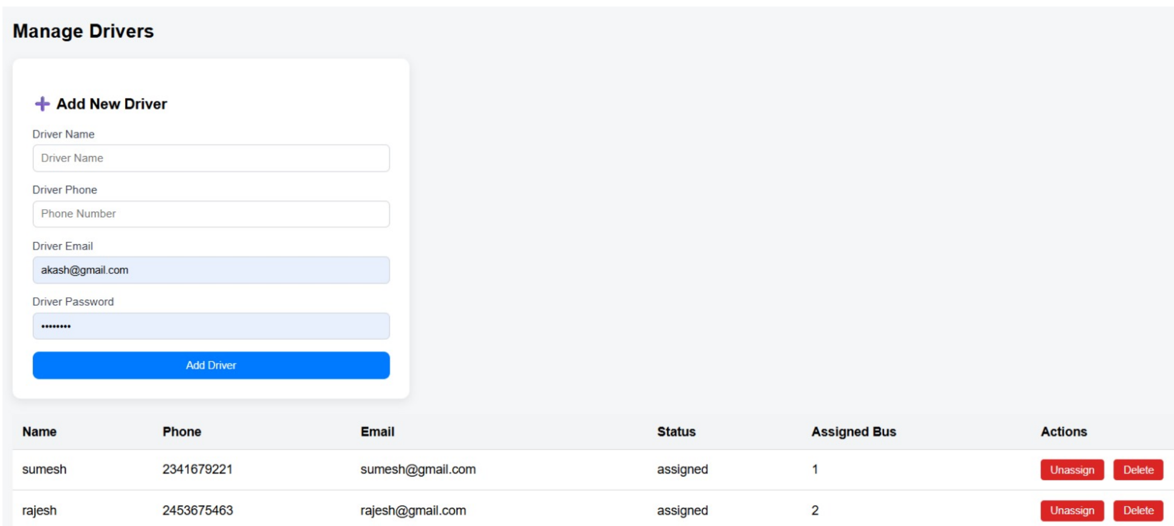

Fig. 8.14: Admin Interface - Manage Drivers

8.1.3 Parent Portal

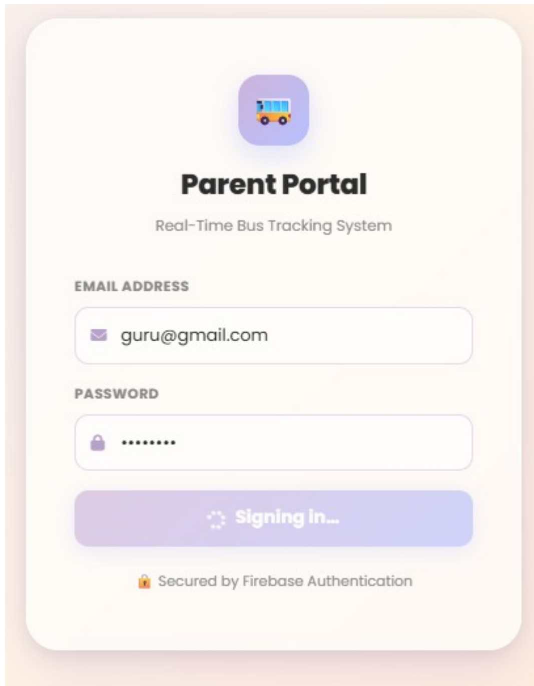

Fig. 8.15: Parent Portal - Parent Login

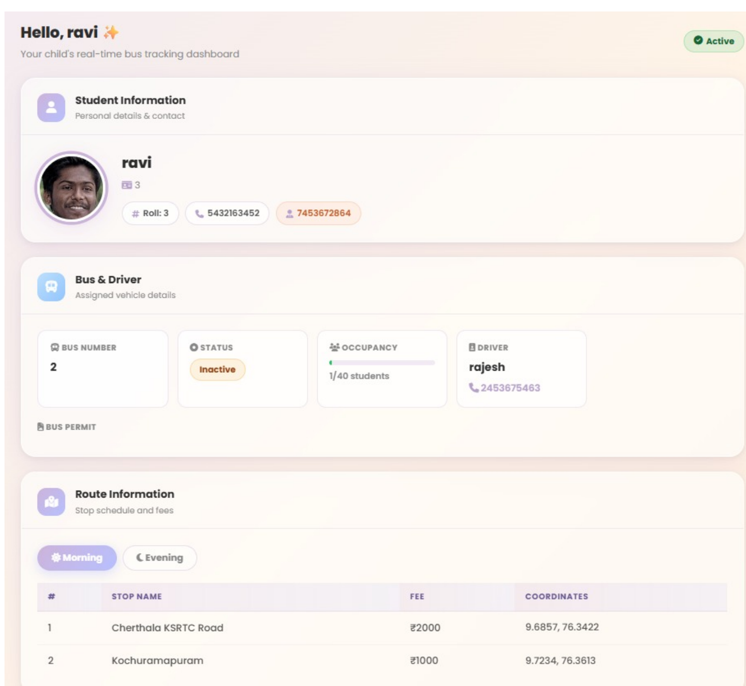

Fig. 8.16: Parent Portal - Dashboard

8.1.4 Driver App

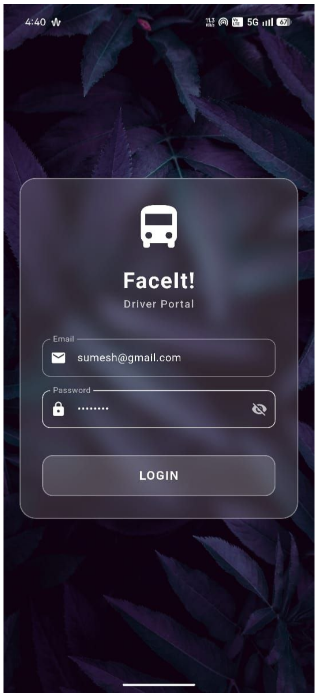

Fig. 8.17: Driver Login

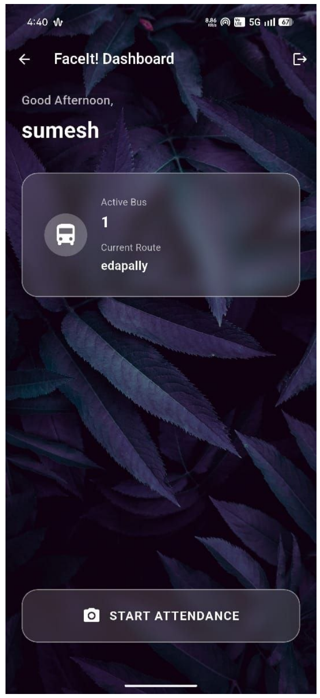

Fig. 8.18: App Dashboard

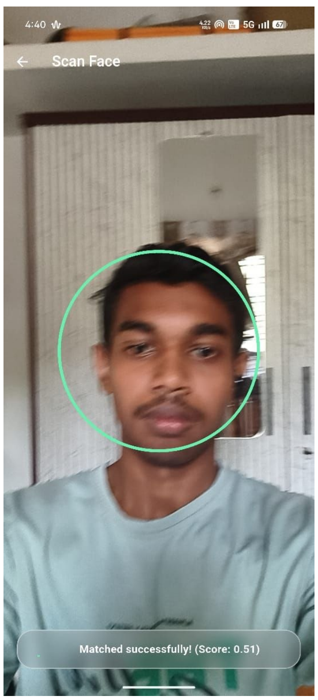

Fig. 8.19: Face Recognition

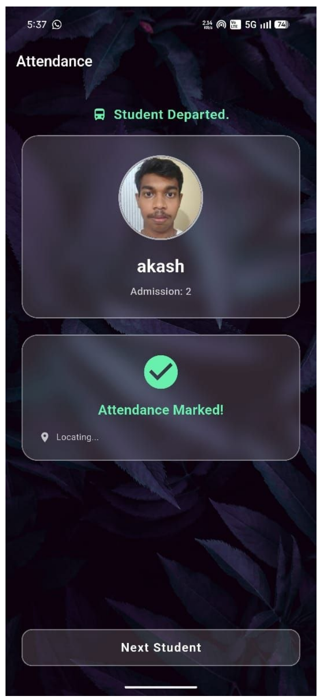

Fig. 8.20: Face Recognition Output

Chapter 9

SAMPLE CODES

9.1 UI State Representation (App)

```
Positioned(
  bottom: 40,
  child: BackdropFilter(
    filter: ImageFilter.blur(sigmaX: 12, sigmaY: 12),
    child: Container(
      child: Row(
        children: [
          if (_isProcessingFace)
            CircularProgressIndicator(),
          Text(_scanStatus),
        ],
      ),
    ),
  ),
);
```

9.2 Real-Time Face Alignment and Detection (App)

```
final List<Face> faces = await _faceDetector.processImage(inputImage);

if (faces.isNotEmpty) {
  double faceCenterX = left + width / 2;
  double faceCenterY = top + height / 2;

  double screenCenterX = _screenSize.width / 2;
  double screenCenterY = _screenSize.height / 2;

  double distance = sqrt(
    pow(faceCenterX - screenCenterX, 2) +
    pow(faceCenterY - screenCenterY, 2)
  );

  if (distance < 120) {
    isCentered = true;
  }
}
```

9.3 Neural Network Face Embedding Generation (App)

```
img.Image resizedImage = img.copyResize(croppedImage, width: 112, height: 1

var input = _imageToByteListFloat32(resizedImage, 112, 127.5, 127.5);
var output = List<double>.filled(192, 0).reshape([1, 192]);

_interpreter.run(input, output);

return List<double>.from(output[0]);
```

```
double cosineSimilarity(List<double> A, List<double> B) {
  double dot = 0, normA = 0, normB = 0;

  for (int i = 0; i < A.length; i++) {
    dot += A[i] * B[i];
    normA += A[i] * A[i];
    normB += B[i] * B[i];
  }
  return dot / (sqrt(normA) * sqrt(normB));
}
```

9.4 Bus Tracking (Web)

```
@bus_bp.route("/api/buses/<bus_id>/tracking")
def api_bus_tracking(bus_id):
    data = tracking_ref.get()

    return jsonify({
        "latitude": data.get("latitude", 0),
        "longitude": data.get("longitude", 0),
        "speed": data.get("speed", 0),
        "attendance": attendance_data,
    })
```

9.5 Bus Enrollment (Web)

```
bus_data = {
    "name": name,
    "bus_number": bus_number,
```

```
    "student_limit": int(student_limit),
    "status": "inactive",
    "permit_photo": permit_path,
}
bus_id = create_bus(db, bus_data)
```

9.6 Student Enrollment (Web)

```
face = detect_and_crop_face(file_path)

embedding = generate_embedding_from_face(face)

student_data = {
    "name": name,
    "embedding": embedding.tolist(),
    "bus_id": bus_id,
    "routes": routes_data,
}
```

9.7 Crop, Embedding Generation, Base64 (Web)

```
results = face_detector.process(img_rgb)

if not results.detections:
    return None

face = img[y1:y2, x1:x2]

embedding = interpreter.get_tensor(output_details[0]['index'])[0]
embedding = embedding / np.linalg.norm(embedding)
```

```
ok, buf = cv2.imencode('.jpg', img)
b64 = base64.b64encode(buf.tobytes()).decode('utf-8')

return f"data:image/jpeg;base64,{b64}"
```

9.8 Fee Generation (Web)

```
if today.month == 1:
    cycle = "JAN"
else:
    cycle = "JULY"

fee_data = {
    "studentId": student_id,
    "amount": student.get("total_fee", 0),
    "cycle": cycle,
}

db.collection("fees").add(fee_data)
```

Chapter 10

FUTURE SCOPE

Building upon the successful implementation of the Smart Bus Tracking and Monitoring System, several enhancements can be incorporated in the future to improve functionality, scalability, and user experience.

- **Mobile Application for Parents:** Develop a dedicated mobile application for parents to track bus location in real time and receive instant notifications when students board or exit the bus.

- **Improved Face Recognition Accuracy:** Enhance the face recognition system to perform efficiently under varying conditions such as low lighting, different angles, and partial occlusions (e.g., masks).

- **Real-Time Alerts and Notifications:** Implement automated SMS or push notifications to inform parents and administrators about student boarding, exit events, and route updates.

- **Integration with College ERP Systems:** Connect the system with existing college management or ERP systems for seamless data synchronization and centralized control.

- **Multi-Bus and Large-Scale Deployment:** Extend the system to support multiple buses and large-scale deployment across institutions with efficient data handling.

- **Offline Functionality:** Enable temporary offline data storage and synchronization once internet connectivity is restored.

- **Enhanced Security Features:** Incorporate encryption and authentication mechanisms to ensure secure handling of student data and prevent unauthorized access.

Overall, the system can be further developed into a comprehensive and scalable smart transportation solution that enhances student safety, operational efficiency, and real-time monitoring capabilities.

Chapter 11

CONCLUSION

The Smart Mobile-Based College Bus Surveillance System has been successfully developed to automate student attendance and enhance safety in college transportation. The system integrates face recognition and GPS technologies into a single mobile platform, enabling real-time identification of students and continuous monitoring of bus movement.

The use of face recognition ensures accurate and contactless attendance marking, eliminating manual errors and preventing proxy attendance. The integration of GPS allows precise tracking of the bus and recording of student boarding and exit locations, improving transparency and monitoring efficiency.

The system also provides effective data management through a centralized database, allowing secure storage and easy access to attendance records and location data. In addition, the ability to detect unregistered individuals enhances overall security and accountability.

The proposed solution is cost-effective and practical, as it requires only a smartphone with built-in camera and GPS, eliminating the need for additional hardware. This makes the system suitable for real-world implementation in educational institutions.

Thus, the system meets its intended objectives by providing a reliable, efficient, and user-friendly solution for smart bus monitoring and student safety.

Chapter 12

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